MOHAMED AYMANE BADIA

Data Science & Machine Learning | Final-year AI & Data Science engineering student (EMSI Marrakech)

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PROFILE

Final-year engineering student in Artificial Intelligence and Data Science (EMSI Marrakech). End-of-year internship (PFA) at Lama Healthcare (USA) on a healthcare provider credentialing platform: front-end lead (Next.js, TypeScript) and document-extraction experiments (LoRA fine-tuning of an LLM, structured JSON output). Personal projects in Deep Learning, NLP and OCR (PyTorch, Hugging Face), with measured results. Seeking a 6-month final-year internship (PFE) in Data Science / Artificial Intelligence, starting February 2027.

PROFESSIONAL EXPERIENCE

PFA Intern – Front-end Development & AI | July 2026 – September 2026
Lama Healthcare (Practice One platform), Las Vegas, Nevada, USA

- Rebuilt, in a team of three, a SaaS credentialing platform for US healthcare providers, within a HIPAA scope.
- Front-end lead: design system and reusable components; delivered the dashboard, provider files, credentialing workflow, alerts, audit log and a public document-upload page.
- Document extraction: compared vision-language models, then fine-tuned Qwen3-1.7B with LoRA on a T4 GPU: 100% valid JSON (60/60) and 87% correct fields (124/143) on the public CORD dataset, 3.3 GB peak VRAM.
- Data security: contributed to role-based access control (RBAC), data masking and access traceability; 6 of 6 roles validated by access tests.
- **Tech stack:** Next.js, TypeScript, Tailwind CSS, Supabase, PostgreSQL, Zod, Hugging Face Transformers, PEFT/LoRA, TRL, Git; English-speaking team.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C#, PHP
Machine Learning: Scikit-learn, Pandas, NumPy, TF-IDF, KMeans, Random Forest, XGBoost/LightGBM, model evaluation
Deep Learning, NLP & OCR: PyTorch, TensorFlow/Keras, CNN, Hugging Face Transformers, PEFT/LoRA, TRL, spaCy, NLTK, Tesseract
Data: PostgreSQL, Supabase, MySQL, Matplotlib, Plotly
Web & APIs: Next.js, React, Tailwind CSS, Flask, ASP.NET Core MVC, EF Core, Laravel, REST APIs, JWT
Tools & practices: Git, Docker, Streamlit, Google Colab, Kaggle, Postman, pytest, MVC / N-Tier architectures

PROJECTS

Invoice Intelligence AI – invoice extraction with OCR + LLM | Python, Transformers, PEFT/LoRA, Tesseract, PyMuPDF, Pydantic, pytest, Streamlit

- Photo/PDF pipeline → Tesseract OCR → Qwen2.5-0.5B-Instruct fine-tuned with LoRA (8.8M trainable parameters out of 503M) → structured JSON → arithmetic repair and Pydantic validation → Streamlit demo.
- Field accuracy of 0.99 vs 0.54 for the base model and 0.53 for a rule-based method (synthetic test set); generalisation limits on real receipts measured and documented.

TrendRadar – Reddit trend detection | Python, spaCy, NLTK, Scikit-learn, XGBoost/LightGBM, Plotly, Streamlit

- End-to-end application (~3,300 lines of Python, 5 modules): Reddit collection with a Parquet cache, French/English NLP (language detection, lemmatization, named entities), TF-IDF vectorization.
- KMeans topic clustering and Random Forest classification, sliding-window burst detection, gradient-boosting virality prediction, interactive dashboard.

Deployed CIFAR-10 image classifier | PyTorch, CUDA (T4 GPU), Streamlit, Docker Compose

- 3-block CNN (Conv2D, BatchNorm, Dropout, ~500k parameters) trained with Adam and StepLR: about 78% test accuracy.
- Streamlit app deployed online and containerized (Docker Compose), with an out-of-domain image filter (MobileNetV2) and rejection of uncertain predictions (confidence and entropy thresholds).

EDUCATION & CERTIFICATIONS

Engineering degree (MEng) in Artificial Intelligence and Data Science | 2022 – 2027
EMSI – Moroccan School of Engineering Sciences, Marrakech

- 5-year engineering program (Master of Science equivalent). Core courses: Machine Learning, Deep Learning, NLP, databases, software engineering.
- **Certification:** Software Design & Project Management (Coursera)

LANGUAGES

Arabic: native | French: bilingual | English: upper-intermediate (B2)